

Virtual LANs (VLANs) – Detailed Theory

What is a VLAN?

A **Virtual Local Area Network (VLAN)** is a logical subdivision of a physical LAN at Layer 2 (Data Link layer) that groups devices into separate broadcast domains regardless of their physical location. VLANs allow network administrators to segment a LAN into smaller parts for better performance, security, and management.

Why VLANs?

- **Segmentation:** VLANs break a single broadcast domain into multiple smaller ones, reducing broadcast traffic and collisions.
- **Security:** Devices on different VLANs cannot directly communicate without routing, which enhances security boundaries.
- **Flexibility:** VLANs allow grouping devices logically even if they are on different physical switches or locations.
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Traffic Management: VLANs help organize network traffic by department, function, or project.

VLAN Concepts

- **Broadcast Domain:** A VLAN defines a broadcast domain; broadcasts sent within a VLAN stay confined to that VLAN.
 - **Trunk Links:** Links between switches that carry traffic for multiple VLANs, typically using tagging protocols like IEEE 802.1Q.
 - **Access Ports:** Ports assigned to a single VLAN; devices connected to these ports are members of that VLAN.
 - **Native VLAN:** The VLAN that carries untagged traffic on a trunk link (default VLAN 1 on Cisco switches).
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VLAN Tagging

- **IEEE 802.1Q Standard:** Adds a 4-byte tag in Ethernet frames to identify the VLAN the frame belongs to when traveling across trunk links.
 - **Tag Structure:** The tag contains a VLAN ID (12 bits), allowing up to 4094 VLANs.
 - **Native VLAN Traffic:** Traffic in the native VLAN is sent untagged on trunk links for backward compatibility.
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VLAN Types

- **Port-based VLAN:** Ports are assigned to VLANs manually.
- **Protocol-based VLAN:** VLAN membership is based on protocols (e.g., IP, IPX).
- **MAC-based VLAN:** Devices are assigned VLANs based on their MAC addresses.
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Dynamic VLAN: VLAN membership assigned dynamically using VLAN Membership Policy Server (VMPS).

VLAN Operation

- When a frame arrives at a switch port:
 - If it is an **access port**, the switch assigns the frame to that port's VLAN.
 - If it is a **trunk port**, the VLAN tag in the frame header is used to forward it correctly.
 - Switches maintain separate MAC address tables for each VLAN.
 - Devices in different VLANs cannot communicate unless routed through a Layer 3 device (router or Layer 3 switch).
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Inter-VLAN Routing

- Since VLANs are separate broadcast domains, a router or Layer 3 switch is required to route traffic between VLANs.
 - **Router-on-a-stick:** A router with a single physical interface configured with multiple sub-interfaces, each mapped to a VLAN.
 - **Layer 3 Switches:** Switches with routing capabilities to route between VLANs internally, improving performance.
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VLAN Management Protocols

- **VLAN Trunking Protocol (VTP):** Cisco proprietary protocol that manages VLAN configuration and propagation across switches in the same VTP domain.
- **Dynamic Trunking Protocol (DTP):** Cisco proprietary protocol to negotiate trunking on a link automatically.
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VTP Modes:

- Server: Can create, modify, delete VLANs.
- Client: Receives VLAN info but cannot modify.
- Transparent: Does not participate in VTP, forwards VTP info.

VLAN Security Considerations

- **VLAN Hopping:** Attack where a device sends double-tagged frames to gain access to VLANs it shouldn't.
- **Native VLAN Security:** Native VLAN should be changed from default VLAN 1 to avoid security risks.
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Port Security: Limits devices by MAC address on ports to prevent unauthorized access.

- **Private VLANs:** VLANs that isolate ports within the same VLAN for enhanced security.

Summary of VLAN Benefits

Benefit	Explanation
Traffic Isolation	Reduces broadcast domains for efficiency
Security	Limits user groups and access
Flexibility	Logical grouping of devices
Scalability	Easier to add or move devices
Network Management	Simplifies changes and troubleshooting